

Dot Product

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3 = |\vec{a}| |\vec{b}| \cos \theta$$

Angle: $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$

Scalar projection: $\text{comp}_{\vec{a}} \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$

Vector projection: $\text{proj}_{\vec{a}} \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|^2} \vec{a}$

Cross Product

$$\vec{a} \times \vec{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta$$

Area of parallelogram = $|\vec{a} \times \vec{b}|$

Area of triangle = $\frac{1}{2} |\vec{a} \times \vec{b}|$

Lines & Planes

Line through P_0 with direction $\vec{v} = \langle a, b, c \rangle$:

$$\vec{r} = \vec{r}_0 + t\vec{v}$$

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct$$

Plane with normal $\vec{n} = \langle a, b, c \rangle$:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

Distance from point to plane:

$$D = \frac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}$$

Vector Functions (Ch. 13)

$$\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$$

Derivative: $\vec{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle$

Unit tangent: $\vec{T}(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}$

Arc length: $L = \int_a^b |\vec{r}'(t)| dt$

Curvature: $\kappa = \frac{|\vec{r}' \times \vec{r}''|}{|\vec{r}'|^3}$

Partial Derivatives (Ch. 14)

Tangent plane at (x_0, y_0, z_0) :

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Linearization:

$$L = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

Total differential: $dz = f_x dx + f_y dy$

Chain Rule

If $z = f(x, y)$, $x = x(t)$, $y = y(t)$:

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

If $x = x(s, t)$, $y = y(s, t)$:

$$\frac{\partial z}{\partial s} = f_x \frac{\partial x}{\partial s} + f_y \frac{\partial y}{\partial s}$$

Implicit: If $F(x, y, z) = 0$:

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$$

Directional Derivative & Gradient

$$\nabla f = \langle f_x, f_y \rangle \quad (\text{or } \langle f_x, f_y, f_z \rangle \text{ in 3D})$$

$$D_{\vec{u}} f = \nabla f \cdot \vec{u} \quad (\vec{u} \text{ is a unit vector})$$

Max rate of change = $|\nabla f|$, in direction ∇f
 ∇f is $\perp$ to level curves/surfaces.

Second Derivative Test

At a critical point (a, b) where $f_x = f_y = 0$:

$$D = f_{xx} f_{yy} - (f_{xy})^2$$

$$D > 0, f_{xx} > 0 \quad \text{local min}$$

$$D > 0, f_{xx} < 0 \quad \text{local max}$$

$$D < 0 \quad \text{saddle point}$$

$$D = 0 \quad \text{inconclusive}$$

Lagrange Multipliers

Optimize f subject to $g(x, y) = k$:

$$\nabla f = \lambda \nabla g \quad \text{and} \quad g = k$$

System: $f_x = \lambda g_x, \quad f_y = \lambda g_y, \quad g = k$

Trig Identities

Pythagorean:

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

Half-Angle (Power Reduction):

$$\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$$

$$\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$$

Double-Angle:

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

Integration Formulas

$$\int \sec \theta \, d\theta = \ln |\sec \theta + \tan \theta| + C$$

$$\int \csc \theta \, d\theta = -\ln |\csc \theta + \cot \theta| + C$$

$$\int \tan \theta \, d\theta = \ln |\sec \theta| + C$$

$$\int \frac{du}{a^2 + u^2} = \frac{1}{a} \arctan \frac{u}{a} + C$$

$$\int \frac{du}{\sqrt{a^2 - u^2}} = \arcsin \frac{u}{a} + C$$

Trig powers:

$$\int \sin^2 \theta \, d\theta = \frac{\theta}{2} - \frac{\sin 2\theta}{4} + C$$

$$\int \cos^2 \theta \, d\theta = \frac{\theta}{2} + \frac{\sin 2\theta}{4} + C$$

$$\int \sec^3 \theta \, d\theta = \frac{1}{2}(\sec \theta \tan \theta + \ln |\sec \theta + \tan \theta|) + C$$

Trig Substitution

See	Let	Identity
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	$1 - \sin^2 = \cos^2$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$	$1 + \tan^2 = \sec^2$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$	$\sec^2 - 1 = \tan^2$

Integration by Parts

$$\int u \, dv = uv - \int v \, du$$

LIATE priority: Logs, Inverse trig, Algebraic, Trig, Exponential.

Coordinate Systems

Polar (r, θ) :

$$x = r \cos \theta, \quad y = r \sin \theta$$

$$r^2 = x^2 + y^2, \quad dA = r \, dr \, d\theta$$

Cylindrical (r, θ, z) :

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z$$

$$dV = r \, dz \, dr \, d\theta$$

Spherical (ρ, ϕ, θ) :

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

$$\rho^2 = x^2 + y^2 + z^2$$

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

$$\phi \in [0, \pi] \text{ from } +z\text{-axis, } \theta \in [0, 2\pi]$$

Multiple Integrals

Surface area of $z = f(x, y)$:

$$A = \iint_D \sqrt{1 + f_x^2 + f_y^2} \, dA$$

Center of mass (ρ = density):

$$\bar{x} = \frac{\iint_D x \rho \, dA}{\iint_D \rho \, dA}, \quad \bar{y} = \frac{\iint_D y \rho \, dA}{\iint_D \rho \, dA}$$

Vector Calc Operations

$$\vec{F} = \langle P, Q, R \rangle$$

$$\text{Gradient: } \nabla f = \langle f_x, f_y, f_z \rangle$$

$$\text{Divergence: } \nabla \cdot \vec{F} = P_x + Q_y + R_z$$

Curl:

$$\nabla \times \vec{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ P & Q & R \end{vmatrix}$$

$$= \langle R_y - Q_z, P_z - R_x, Q_x - P_y \rangle$$

$$\text{Conservative test: } \nabla \times \vec{F} = \vec{0}$$

Line Integrals

$$\text{Arc length: } \int_C ds = \int_a^b \|\vec{r}'(t)\| \, dt$$

$$\text{Work: } \int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) \, dt$$

Fund. Thm. of Line Integrals:

$$\text{If } \vec{F} = \nabla f: \quad \int_C \vec{F} \cdot d\vec{r} = f(B) - f(A)$$

Surface Integrals

Flux through $z = f(x, y)$ (upward normal):

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_D \vec{F} \cdot \langle -f_x, -f_y, 1 \rangle \, dA$$

$$\text{Parametric surface: } d\vec{S} = (\vec{r}_u \times \vec{r}_v) \, du \, dv$$

Surface area (parametric):

$$A = \iint_D \|\vec{r}_u \times \vec{r}_v\| \, du \, dv$$

The Big Three Theorems

Green's Theorem (2D, CCW):

$$\oint_C P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \, dA$$

Stokes' Theorem (surface $\rightarrow$ boundary):

$$\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \oint_C \vec{F} \cdot d\vec{r}$$

Divergence Theorem (solid $\rightarrow$ boundary):

$$\iiint_E (\nabla \cdot \vec{F}) \, dV = \iint_S \vec{F} \cdot d\vec{S}$$